Heuristic Evaluation

Structure of the individual report

Part I: Your Name

Francesco Anzoino, s304125.

Part II: Project Description

Smartwatch application for novice gym-goers. The purpose is to guide individuals in the practice of gym exercises, ensuring proper execution. It provides support for organizing the structure of the gym exercise circuit and encourages regular attendance at the gym.

Part III: Evaluation Execution

The evaluation was conducted in person using a printed paper prototype. Two facilitators assisted in the evaluation, one providing a brief description of the application's functionality and taking notes, while the other facilitated the replacement of application views to simulate the usage flow.

The tasks to be performed were as follows:

- Task “Simple”: Learn how to perform an exercise.
- Task “Moderate: Replace an exercise with another one related to the same muscle group.
- Task “Complex”: Reach a certified level.

The evaluation proceeded in the following order:

1. Brief overview of the project and the purpose of the smartwatch application
2. Articulation of tasks to be performed, ranging from simple to complex.
3. Free execution of the three tasks. Once one task was completed, the next one of greater difficulty was explained, allowing freedom in its execution.

The materials provided at the end of the evaluation include the PDF of the paper prototype and a recording made during the evaluation itself.

Part IV: List of Violations

1. H1 Visibility of system status
   Where: "Muscle Groups" View
   What: Clicking the "+" button does not provide any notification.
   Why: It is not possible to understand what happens to the application's status once that muscle group is added, including where and how it is added.
   Severity: 3

2. H8 Recognition rather than recall
   Where: "Specific Muscle Group" View
   What: The "+" button is redundant.
   Why: Adding an exercise is possible even after viewing the specific exercise. Therefore, inserting it beforehand is unnecessary and misleading compared to clicking on the exercise name to view it.
   Severity: 2

3. H3 User control and freedom
   Where: “Exercise Countdown” View
   What: A countdown is automatically initiated.
   Why: Adding an automatic countdown and starting the time count automatically restricts the user's freedom to begin according to their preferences.
   Severity: 3

4. H1 Visibility of system status
   Where: “Exercise in Progress” View
   What: Presence of a number and a timer without an explanation of their purpose.
   Why: The user cannot understand what the increasing number refers to. Additionally, the running timer is not clearly indicated.
   Severity: 2

5. H7 Flexibility and efficiency of use
   Where: “Exercise Completed” View
   What: Lack of a shortcut to return to the main view with muscle groups.
   Why: To return to the initial view after completing an exercise, the user must scroll back through the numerous navigated views, indicating the absence of a convenient shortcut.
   Severity: 2

6. H5 Error prevention
   Where: “Plan” View
   What: Absence of a method to delete an inserted exercise.

Why: The user lacks the ability to remove an exercise added to their plan, for example, if it was included by mistake.
Severity: 2

7. H3 User control and freedom
   Where: “Plan” View
   What: Absence of a method to delete an inserted exercise.
   Why: The user lacks the ability to remove an exercise added to their plan, for example, if it was included by mistake.
   Severity: 2

8. H10 Help and documentation
   Where: “Plan” View
   What: It's not clear what happens when clicking the "change" button.
   Why: The term "change" could imply either a user-initiated choice or a random switch of the exercise upon clicking the "change" button.
   Severity: 2

9. H3 User control and freedom
   Where: “Plan” View
   What: Clicking the "change" button randomly switches the exercise.
   Why: The exercise replacement is random upon clicking the button and is not user-initiated. Additionally, there is no option to revert to the previous exercise after it has been changed.
   Severity: 3

10. H10 Help and documentation
    Where: “Quiz” View
    What: “Lv” text
    Why: The meaning and method to increase the level indicated by the text "Lv" are not clear.
    Severity: 2

11. H6 Help and documentation
    Where: “Quiz” View
    What: “Quiz” text
    Why: The term "Quiz" does not provide clarity on the specific muscle group to which the selected quiz corresponds.
    Severity: 2

12. H10 Help and documentation
    Where: “Quiz” View
    What: “60:00/80:00” text

Why: The meaning of the text "60:00/80:00" is unclear, and it's uncertain whether it denotes remaining hours or some other information.
Severity: 2

13. H3 User control and freedom
    Where: "Quiz Selected" View
    What: Lack of a button to finish the quiz early.
    Why: There is no option to conclude a quiz before its expected completion time by returning to the view with all available quizzes.
    Severity: 2

14. H2 Match between system and the real world
    Where: “Quiz” View
    What: The presence of a selectable checkbox next to a quiz.
    Why: Clicking on this checkbox leads me to take a quiz, but I would expect to mark the quiz as completed, given the presence of a selectable checkbox as an input.
    Severity: 2

15. H2 Match between system and the real world
    Where: “Quiz Completed” View
    What: The presence of the arrow in the top left corner.
    Why: With this arrow, the user expects to return to the view of the individual question in the quiz; however, the arrow takes them back to the screen with all selectable quizzes.
    Severity: 3

16. H1 Visibility of system status
    Where: “Quiz” View
    What: The ability to access a completed quiz again.
    Why: There is no need to access and retake a quiz that has already been completed, unless there is a clear indication of the opportunity to improve one's previously achieved maximum score.
    Severity: 2

17. H4 Consistency and standards
    Where: “Quiz Completed” View
    What: The presence of the arrow in the top left corner.
    Why: The arrow directly takes you back to the main quiz view, unlike the view with recorded times for individual exercises where the top-left arrow does not lead to the view with the selection of muscle groups.
    Severity: 3

Part V: Summary and Recommendations

Report in the table below the total number of identified violations.

Heuristic	**# violations**
H1: Visibility of system status	3
H2: Match between system and the real world	2
H3: User control and freedom	4
H4: Consistency and standards	1
H5: Error prevention	1
H6: Recognition rather than recall	1
H7: Flexibility and efficiency of use	1
H8: Aesthetic and minimalist design	1
H9: Help users recognize, diagnose, and recover from errors	0
H10: Help and documentation	3
HN: Non-heuristic issue	0

The prototype is certainly functional, and completing tasks was easy but not entirely intuitive. There was some confusion in certain sections, such as the 'Plan' and 'Quiz' sections, where a lack of overall control over free modification of the plan was observed and clarity in the interface was lacking in some on-screen texts. Specifically, the logic behind replacing an exercise for task 2 is not quite clear and allows for some freedom in its execution, thereby adding a challenge to the completion of the related task. Positive aspects include the few on-screen elements that do not create confusion. Recommendations include providing clearer user guidance in the app sections and improved clarity on the meaning of on-screen elements and the outcomes of certain interactions (e.g., adding an exercise to the plan).

A3 : HEURISTIC EVALUATION

Part I: Your Name

Gravile Matteo, student's ID: s322994

Part II: Project Description

The project I will analyze involves creating a smartwatch application named GYM TONIC focused on 'Education & Learning', aiming to guide beginners in the gym by explaining exercises, offering real-time support, and concluding with quizzes for certification.

Part III: Evaluation Execution

The heuristic evaluation took place in person during the project session, where I collaborated with two team members. Throughout the evaluation I kept paper notes which, combined with the video documentation, proved essential for this work.
I started the evaluation together with a team member who acted as a "facilitator", playing a crucial role in introducing the purpose of the application, outlining the three designated evaluation tasks and providing valuable insights into smartwatch technology, an area that I found challenging. At the same time, the other team member acted as a "computer", walking me through the use of the application, presenting relevant screens after my interactions. Thus arriving at the completion of the activities and an exploration of the other functions. Initially, interactions with the facilitator were frequent, especially when dealing with screens that posed challenges for a user unfamiliar with smartwatch technology.

Part IV: List of Violations

1. **HN: Non-heuristic issue**
   Where: Main page
   What: Upon opening the application, there is a lack of clarity regarding the user's location
   Why: The initial interface presents only a list of exercises without a clear title, making it challenging for users to discern their location and purpose. Using the "Plan" page as the starting point could enhance user orientation
   Severity: 2

2. **H1 Visibility of system status**
   Where: Page with timer and number of repetitions
   What: The option to go back is available, but the exercise progress indicator is no longer visible.
   Why: By changing the screen and continuing to train I have no feedback on the exercise and I can no longer keep track of the number of repetitions and time
   Severity: 2

3. **H1 Visibility of system status**
   Where: Page of the technical sheet of the exercise
   What: Adding an exercise to the plan lacks feedback on the action's success
   Why: Clicking the + the user does not have feedback on what happened but will only find the exercise on the "plan" page when he has to train
   Severity: 1

4. **H2 Match between system and the real world**
   Where: Plan page
   What: The Plan Title
   Why: The term "Plan" is too generic, failing to convey the specific content of the page. A more descriptive title like "MY TRAINING" would enhance clarity
   Severity: 1

5. **H4: Consistency and standards**
   Where: Page of the technical sheet of the exercise
   What: The Play button
   Why: The generic "Play" button suggests video playback but initiates a countdown for the timer. A more fitting label like "START" would align better with user expectations
   Severity: 2

6. **H3 User control and freedom**
   Where: Page with timer and number of repetitions
   What: Lack of a pause button during exercises
   Why: The absence of a pause button restricts user control, forcing them to either complete the exercise or return to the exercise sheet, resulting in potential loss of progress
   Severity: 3

7. **H3 User control and freedom**
   Where: Plan page
   What: The exercise change button lacks user choice options
   Why: Clicking the button automatically alters the exercise with no undo or redo functionality and no freedom for users to make a personalized choice
   Severity: 4

8. **H7: Flexibility and efficiency of use**
   Where: Plan page
   What: Absence of the option to delete an exercise directly
   Why: The only way to delete exercises from the plan is through the exercise sheet, requiring additional steps. Including a shortcut to delete exercises directly from the plan page would enhance efficiency
   Severity: 2

9. **H7: Flexibility and efficiency of use**
   Where: Page with the summary of the exercise
   What: Statistics are shown post-exercise but cannot be revisited
   Why: After completing an exercise, accessing the statistics page requires restarting the same exercise. Providing a direct path to review statistics would improve user control
   Severity: 2

10. **H3 User control and freedom**
    Where: Quiz page
    What: Inability to view incorrect answers after completing the quiz
    Why: Users lack the opportunity to learn from mistakes as there is no option to review incorrect answers or retake the quiz
    Severity: 3

11. **H5: Error prevention**
    Where: Quiz page
    What: No option to cancel starting a quiz
    Why: Clicking on the quiz initiates it without a confirmation prompt, leading to the possibility of unintentional quiz starts. Incorporating a confirmation step would prevent such errors
    Severity: 4

12. **H4: Consistency and standards**
    Where: TIME in Page with timer and number of repetitions and in QUIZ page
    What: Inconsistent time display formats
    Why: The time is presented differently in the two contexts, causing confusion. Standardizing the time display format would enhance clarity
    Severity: 2

13. **H5: Error prevention**
    Where: Plan page
    What: The switch button activates without confirmation
    Why: Accidental clicks may result in the loss of an exercise without a confirmation prompt. Implementing a confirmation step would prevent such errors
    Severity: 4

14. **H5: Error prevention**
    Where: Quiz page
    What: Inability to change an answer after submission
    Why: By unintentionally answering a question, the quiz immediately moves on to the next one without asking the user for confirmation. I would help prevention for the user by giving the ability to change the answer of each quiz at any time by adding a "SEND" button to the last question of the quiz. By doing this, only when the user is sure to send the answers does he click on the button
    Severity: 3

15. **H8: Aesthetic and minimalist design**
    Where: Main page and Quiz page
    What: Exercise and quiz buttons lack apparent clickability
    Why: Unlike other buttons that clearly indicate clickability, the exercise and quiz buttons resemble plain text on a menu. Enhancing their visual cues would improve user interaction
    Severity: 1

16. **H8: Aesthetic and minimalist design**
    Where: Quiz app
    What: A checkbox appears upon unlocking the quiz
    Why: It is a minimal design but it is misunderstandable, it is better to use a symbol like a padlock and when it is unlocked it disappears, only after a user has tried the quiz does an X or a V appear based on how it went
    Severity: 2

17. **H10: Help and documentation**
    Where: main page
    What: Lack of information on application functionality
    Why: at the beginning it is very difficult for a user to understand how to use the application and hidden information would greatly help understanding for a more informed use
    Severity: 3

18. **H10: Help and documentation**
    Where: quiz page
    What: Unclear criteria and functionality of quizzes
    Why: Including documentation at the top of the Quiz page would assist users in understanding criteria for unlocking quizzes and interpreting the significance of different levels
    Severity: 3

Part V: Summary and Recommendations

Report in the table below the total number of identified violations.

Heuristic	**# violations**
H1: Visibility of system status	2
H2: Match between system and the real world	1
H3: User control and freedom	3
H4: Consistency and standards	2
H5: Error prevention	3
H6: Recognition rather than recall	0
H7: Flexibility and efficiency of use	2
H8: Aesthetic and minimalist design	2
H9: Help users recognize, diagnose, and recover from errors	0
H10: Help and documentation	2
HN: Non-heuristic issue	1

Overall, I think the application is well designed with a minimalist approach that effectively performs all the basic tasks expected. This achievement is noteworthy given the specific technology for which it was developed and the inherent complexity of the application.
However, I see room for improvement, particularly in improving the user experience. I suggest incorporating documentation or visual aids to help users navigate the application more effectively and understand its functionality without feeling overwhelmed by the extensive functionality.
Once users have a better understanding of the application, I propose to give them more freedom. This includes allowing them to revisit sections where they cannot currently be accessed quickly and providing the ability to review quizzes to identify and understand errors.
Finally, I recommend introducing a profile screen that shows the user's level, a complete overview of the exercises completed over time, showing progress and improvements. Additionally, incorporating a feature to mark exercises as favorites would allow users to prioritize certain exercises for easy access when making choices.